FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – XI

PAPER-2

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
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2.	**A**	25.	**B**	48.	**A**
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5.	**B,C**	28.	**B,C**	51.	**C, D**
6.	**B,C**	29.	**A,C,D**	52.	**A,D**
7.	**C,D**	30.	**B,D**	53.	**A,C,D**
8.	**A,C**	31.	**A,D**	54.	**A,B,D**
9.	**C**	32.	**D**	55.	**A**
10.	**C**	33.	**B**	56.	**B**
11.	**4**	34.	**4**	57.	**2**
12.	**2**	35.	**3**	58.	**1**
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14.	**5**	37.	**2**	60.	**1**
15.	**2**	38.	**3**	61.	**8**
16.	**1**	39.	**1**	62.	**9**
17.	**2**	40.	**6**	63.	**1**
18.	**4**	41.	**7**	64.	**5**
19.	**5**	42.	**6**	65.	**0**
20.	**4**	43.	**5**	66.	**4**
21.	**00000.15**	44.	**00100.00**	67.	**00002.63**
22.	**00126.00**	45.	**00000.66**	68.	**00000.50**
23.	**00001.32**	46.	**00009.87**	69.	**00121.17**

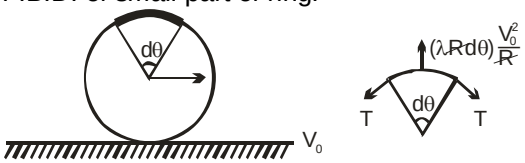
Physics

PART – I

SECTION – A

1. **B**

Sol. F.B.D. of small part of ring.



$$2T\sin\left(\frac{d\theta}{2}\right)=\lambda d\theta v_0^2 \qquad v_{wave}=\sqrt{\frac{T}{\mu}}$$

$$\lambda \cancel{d\theta} v_0^2 = T\cancel{d\theta} \qquad T=\lambda v_0^2=\sqrt{\frac{\cancel{\lambda} v_0^2}{\cancel{\lambda}}}=v_0$$

Time take by hoop in one complete revolution $=\frac{2\pi}{v_0}=\frac{2\pi R}{V_0}$

Distance traveled by wave in this time $=\frac{\cancel{V_0}\times 2\pi R}{\cancel{V_0}}$

$\therefore$ angle rotated $=2\pi$

2. **A**

Sol. Writing energy equation at any instance we get,

$mg\frac{2R}{\pi}(1-\cos\theta)+\frac{1}{2}I\omega^2=\text{constant}$ (where I = moment of inertia about bottom most point)

$$I_{CM}=mR^2-m\left(\frac{2R}{\pi}\right)^2$$

$$I_{\text{Bottom point}}=I_{CM}+m\left(R-\frac{2R}{\pi}\right)^2$$

$$=mR^2-\cancel{m\left(\frac{2R}{\pi}\right)^2}+mR^2+\cancel{m\left(\frac{2R}{\pi}\right)^2}-2.m.R.\frac{2R}{\pi} \qquad I_B=2mR^2-4\frac{mR^2}{\pi}$$

$$mg\frac{2R}{\pi}(1-\cos\theta)+\frac{1}{\cancel{2}}\times\cancel{2}\left(mR^2-\frac{2mR^2}{\pi}\right)\omega^2=\text{constant}$$

Differentiating if we get,

$$mg\frac{\cancel{2}R}{\pi}\cdot\sin\theta\cdot\frac{\cancel{d\theta}}{\cancel{dt}}+\left(mR^2-\frac{2mR^2}{\pi}\right)\cdot\cancel{2}\,\cancel{\omega}\frac{d\theta}{dt}=0$$

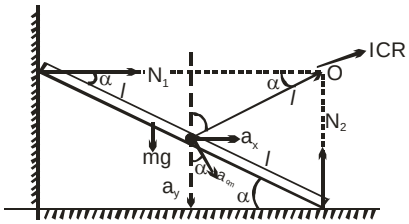
$$mg\frac{R}{\pi}\cdot\theta+mR^2\left(1-\frac{2}{\pi}\right)\frac{d^2\theta}{dt^2}=0$$

$$\frac{d^2\theta}{dt}+\frac{\cancel{m}g\frac{\cancel{R}}{\pi}}{\cancel{m}R^{\cancel{2}}\left(1-\frac{2}{\pi}\right)}\theta=0$$

$$\omega^2=\frac{g}{R(\pi-2)} \qquad \omega=\sqrt{\frac{g}{R(\pi-2)}} \qquad \textbf{Ans.:}\ T=2\pi\sqrt{\frac{R(\pi-2)}{g}}$$

3. **D**

Sol. F.B.D when at any instant rod values angle ' α ' with horizontal when rod leaves contact $N_1 = 0$, $a_x = 0$, $v_x = \max$.



Writing energy equation we get,

$$mgl(\sin\theta_0 - \sin\theta) = \frac{1}{2} I_0 \omega^2$$

$$mgl(\sin\theta_0 - \sin\theta) = \frac{1}{\alpha}\left[\frac{m(2l)^2}{12} + ml^2\right]\omega^2$$

$$= \frac{1}{2}\left[\frac{ml^2}{3} + ml^2\right]\omega^2 \qquad = \frac{1}{2}\left[\frac{4}{3}ml^2\omega^2\right] \quad = \frac{1}{2} \times \frac{4}{3} mv_0^2$$

$$v_{cm} = \sqrt{\frac{3}{2}gl(\sin\theta_0 - \sin\theta)} \qquad v_x = v_{cm}\sin\theta$$

$$v_x = \sqrt{\frac{3}{2}gl}\left(\sqrt{\sin\theta_0 - \sin\theta}\right)\sin\theta \quad \frac{dv_x}{d\theta} = \sqrt{\sin\theta_0 - \sin\theta}\cdot\cos\theta - \sin\theta\frac{1-\cos\theta}{2\sqrt{\sin\theta_0 - \sin\theta}} = 0$$

$$\frac{2(\sin\theta_0 - \sin\theta)\cdot\cos\theta - \sin\theta}{2\sqrt{\sin\theta_0 - \sin\theta}} = 0$$

$$3\sin\theta_0 = 2\sin\theta_0$$

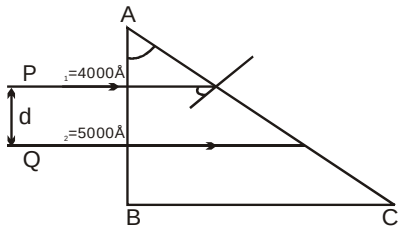
Ans.: $\sin\theta = \frac{2}{3}(\sin\theta_0)$

4. **AB**

Sol. $\sin C$ will be smaller for smaller λ .

For $\left(0 < C < \pi/2\right)$ C will be small for smaller λ .

$\therefore$ TIR will occur for 4000 Å and not 5000 Å



$$\mu = 1.20 + \frac{b}{\lambda^2} \qquad 1.25 = 1.20 + \frac{b}{\lambda^2} \qquad \Rightarrow \frac{b}{\lambda^2} = 0.05$$

$$b = 0.05 \times (4000 \times 10^{-10})^2 \qquad b = 8 \times 10^{-15} m^2$$

From Snell's law

$$\mu_1 \sin i = 1 \sin r \qquad \mu = \frac{\sin r}{\sin i}$$

$$1.0 + \frac{8 \times 10^{-15}}{(5000 \times 10^{-10})^2} = \frac{\sin r}{0.8} \qquad \Rightarrow \sin r = 0.9856 \qquad \Rightarrow r = 80.3^0$$

$$8 = (r - i) = 80.3^0 - 53.1^0$$

$$\delta = 27.2^0$$

5. **BC**

Sol. No. of photons emitted per second $= \frac{3.2 \times 10^{-3}}{\left(\frac{hc}{\lambda}\right)}$

$$= \frac{3.2 \times 10^{-3}}{5 \times 1.6 \times 10^{-19}}$$

$$= \frac{2}{5} \times 10^{16} = 4 \times 10^{15} \text{ photons/sec.}$$

Photons striking the metallic sphere $= \frac{4 \times 10^{15}}{4 \times \pi \times (0.8)^2} \times \pi \times (8 \times 10^{-3})^2$

$$= \frac{10^{15} \times 100}{8 \times 8} \times 64 \times 10^{-6}$$

$$= 10^{11} \text{ photon/sec}$$

$\therefore$ No. of electron emitted per second = $\frac{10^{11}}{10^6} = 10^5$ electron/sec.

K.E_{max} of electron = 5 - 3 = 2eV

Potential at which emitted stops = 2 Volt

$\therefore \frac{Kq}{r} = 2V \qquad q = \frac{2 \times 8 \times 10^{-3}}{9 \times 10^9} = \frac{16}{9} \times 10^{-12} C$

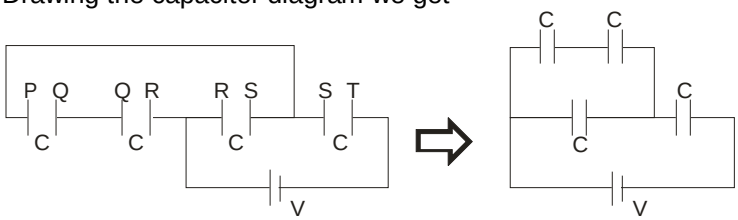
$\therefore$ Time after which sphere acquires this change

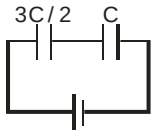
$$= \frac{16}{9} \frac{\times 10^{-12} \times 10}{46 \times 10^{-19} \times 10}$$

$$t = \frac{10^3}{9} = 111.11 \text{ sec.}$$

6. **BC**

Sol. Drawing the capacitor diagram we get





$C = \frac{A\varepsilon_0}{d}$

$\frac{1}{C_{eq.}} = \frac{2}{3C} + C \qquad C_{eq} = \frac{3}{5}C$

After charge distribution on plates we get, Charge on plate P = $\frac{\varepsilon_0 AV}{5d}$ charge on plate

$R = \frac{3\varepsilon_0 AV}{5d}$

7. **CD**

Sol. It is clear from theory of electrostatics (C, D)

8. **AC**

Sol. Flux linked with loop due to its own magnetic field,

$\phi = \frac{\mu_0 I}{2r}(\pi r^2) = \frac{\mu_0 \pi r I}{2}$

emf induced $= -\frac{d\phi}{dt} = \varepsilon = -\frac{\mu_0 \pi r}{2}\frac{dI}{dt}$

$I = \frac{\varepsilon}{R} = -\frac{\mu_0 \pi r}{2R}\frac{dI}{dt} \qquad \int_{I_0}^{I} \frac{dI}{I} = -\int_0^t \frac{2R}{\mu_0 \pi r}.dt; I = I_0 e^{-2Rt/\mu_0 \mu r}$

Now, $10^{-3} I_0 = I_0 e^{-\frac{2Rt}{\mu_0 \pi r}}$ which gives $t = \frac{3\pi^2 \ln 10}{10^{10}} s$.

9. **C**

10. **C**

Sol. **(9 - 10)**

P_0, V_0, T_0	P_0, V_0, T_0	P_0, V_0, T_0
A_1	A_2	A_3

Initial condition

P, V_1, T_1	$P, V, \frac{9T_0}{4}$	$P, V, \frac{9T_0}{4}$
A_1	A_2	A_3

final equilibrium

Considering chamber A_2 and A_3,

No. of moles = constant $\quad \frac{P_0 2V_0}{2RT_0} = \frac{P 2V}{2 \frac{9T_0}{4}}$

$$PV = \frac{9}{4} P_0 V_0$$

Apply adiabatic condition

$P_0(2V_0)^\gamma = P(2V)^\gamma \qquad V = \left|\frac{2}{3}\right|^3 V_0$

$P_0(V_0)^\gamma 2^\gamma = P 2^\gamma v^\gamma \qquad V = \frac{8}{27} V_0$

$P_0 V_0^\gamma = PV^\gamma$

$V^\gamma = \frac{P_0 V_0^\gamma}{P} \qquad V_1 = 3V_0 - 2V = 3V_0 - \frac{16V_0}{27}$

$V^\gamma = \frac{P_0 V_0^\gamma}{\frac{9}{4}\frac{P_0 V_0}{V}} \qquad = \frac{81V_0 - 16V_0}{27}$

$V^{\gamma-1} = \frac{4}{9} V_0^{\gamma-1} \qquad V_1 = \frac{65V_0}{27}$

$V^{2/3} = \frac{4}{9} V_0^{2/3}$ From above volume we can calculate the final pressure.

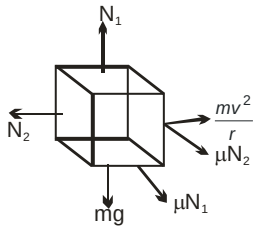
$V^2 = \left(\frac{4}{9}\right)^3 V_0^2$ Final pressure $P = \frac{243}{32} P_0$ and

$V = \frac{4}{9}^{3/2} V_0$ Final temperature $T = \frac{585}{32} T_0$

SECTION – C

11. **4**

Sol. F.B.D of block at any instant



$\mu(N_1 + N_2) = -m\upsilon \frac{d\upsilon}{dx} \qquad \Rightarrow \mu\left(mg + \frac{m\upsilon^2}{r}\right) = -m\upsilon \frac{d\upsilon}{dx} \qquad \Rightarrow \mu\left(\frac{rg + \upsilon^2}{r}\right) = -\upsilon \frac{d\upsilon}{dx}$

$$\int_0^{2\pi}\frac{\mu}{r}dx=\int_0^{2\pi}-\frac{vdv}{(rg+v^2)}dx \qquad \left(rg+v^2=t,\ 2vdv=dt, vdv=\frac{dt}{2}\right)$$

$$\int\frac{\mu}{r}dx=-\int\frac{dt}{t\cdot 2}\ \Rightarrow\ \frac{\mu}{\cancel{r}}\times 2\pi\cancel{r}=-\frac{1}{2}\left[\ln\frac{rg}{rg+v_0^2}\right]$$

$$\mu.4\pi=\ln\left(\frac{rg+v_0^2}{rg}\right) \qquad \Rightarrow rg+v_0^2=rge^{4\pi\mu} \qquad \Rightarrow v_0=\sqrt{rg(e^{4\pi\mu}-1)}$$

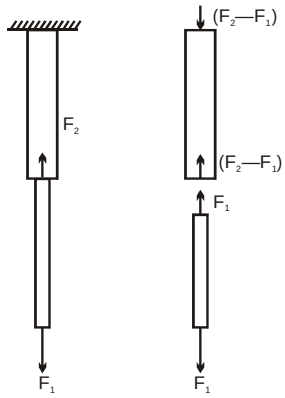
12. **2**

Sol. Deflection of C = 0 $\qquad \frac{F_1.L_2}{A_2Y}=\frac{(F_2-F_1)L_1}{A_1Y}$

$$\frac{F_1L_2}{A_2}+\frac{F_1L_1}{A_1}=\frac{F_2L_1}{A_1}$$

$$\frac{F_2}{F_1}=\left|\frac{\frac{L_2}{A_2}+\frac{L_1}{A_1}}{\frac{L_1}{A_1}}\right|=\left(1+\frac{L_2A_1}{A_2L_1}\right)$$

$$\frac{F_2}{F_1}=2$$



13. **2**

Sol. Consider refraction from oil surface.

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$

$$\frac{7}{\cancel{4}v}+\frac{1}{\cancel{24}_{6}}=\frac{3}{\cancel{4}\times 6}$$

$$\frac{7}{v}=\frac{3}{6}-\frac{1}{6}$$

$$\frac{7}{v}=\frac{1}{3}$$

$v=21$ cm

This will act as an object for oil water interface.

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=0$$

$$\frac{4}{3v}-\frac{7}{4\times 21}=0$$

$$\frac{4}{\cancel{3}v}-\frac{\cancel{7}}{4\times\cancel{21}_{\cancel{3}}}=0 \qquad \Rightarrow v=16$$

$\therefore\ x=18-16=2$

14. **5**

Sol. $\frac{1}{Z_{eq}}=\frac{1}{Z_1}+\frac{1}{Z_2}+\frac{1}{Z_3}$ (for parallel circuit)

$$=\frac{1}{4+2i}+\frac{1}{3-4i}+\frac{1}{8+6i}$$

Solving this we will get $Z_{eq.}=\frac{5}{2}+0i$

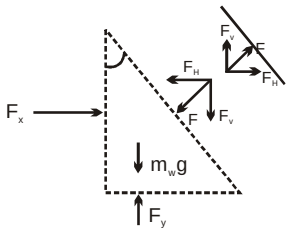
Now, over all impedance of the circuit is $Z=\frac{5}{2}+\frac{5}{2}+5i=5+5i$

Therefore, $|Z|=5\sqrt{2}$

Current supplied by source $=\frac{25\sqrt{2}}{5\sqrt{2}}=5$ **Ans.**

15. **2**

Sol. F.B.D.



$F_x = \rho g \frac{h}{2} \times h \times b$

$F_x = \rho g \frac{h^2}{2} b$

$F_y = \rho g h \times h \tan\theta . b$

$F_y = \rho g h^2 \tan\theta b$

$F_H = F_x$ $\quad F_v = F_y - mg$ $\quad m_w g = \frac{1}{2} \times h \tan\theta . h \times b \times \rho g$

$F_H = \rho g \frac{h^2}{2} b$ $\quad = \frac{1}{2} \rho g h^2 \tan\theta b$ $\quad m_w g = \frac{1}{2} \rho g h^2 \tan\theta u$

Friction force on wedge $= \mu \times F_H = \mu . \rho g \frac{h^2}{2} b$

Max. height of water which can be filled, is the height when wedge starts moving up.

$F_v = mg + \mu F_H \Rightarrow F_v = mg + \mu \rho g \frac{h^2}{2} b \Rightarrow \frac{1}{\cancel{2}} \rho g^2 \tan\theta b = 2mg + \mu \rho g \frac{h^2}{\cancel{\phi}} b$

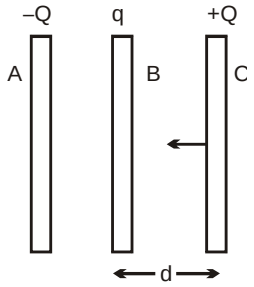
$\Rightarrow h^2 (\rho g \tan\theta - \mu \rho g b) = 2mg$

$h^2 = \frac{2mg}{\rho g \tan\theta b - \mu \rho g b} \Rightarrow h = \sqrt{\frac{2m \cancel{g}}{\cancel{g} \rho b (\tan\theta - \mu)}} \Rightarrow h = \sqrt{\frac{2m}{gb(\tan\theta - \mu)}}$

16. 1

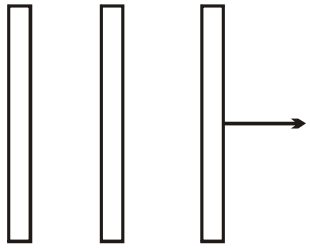
Sol. At any instant force on plate $C = \frac{(Q-q)Q}{2S\varepsilon_0}$

word done = $\frac{(Q-q)Qd}{2S\varepsilon_0}$



After the collision charge on plates A, B, C become

–Q (Q + q/2) (q/2)



Work done $= \frac{(\cancel{Q} + \frac{q}{\cancel{2}} - \cancel{Q}) . \frac{q}{\cancel{2}} d}{2S\varepsilon_0} = \frac{\left[\frac{q}{2}\right]^2 d}{2S\varepsilon_0}$

Total work

done $= \frac{(Q-q)Qd}{2S\varepsilon_0} + \frac{\left(\frac{q}{2}\right)^2 d}{2S\varepsilon_0} = \frac{d}{2S\varepsilon_0}\left[Q^2 - qQ + \left(\frac{q}{2}\right)^2\right]$

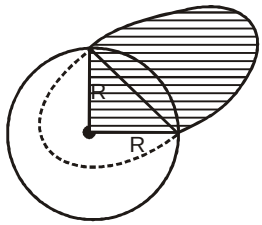
$= \frac{d}{2S\varepsilon_0}\left[\left(Q - \frac{q}{2}\right)^2\right]$

Total work done = charge in K.E.

$\frac{1}{\cancel{2}} mv^2 = \frac{d}{\cancel{2}S\varepsilon_0}\left(Q - \frac{q}{\cancel{2}}\right)^2 \Rightarrow v = \sqrt{\frac{d}{ms\varepsilon_0}}\left(Q - \frac{q}{2}\right) \Rightarrow v = \sqrt{\frac{d}{ms\varepsilon_0}} . \left[\frac{3}{2} - \frac{1}{2}\right] = 1$

17. **2**

Sol. Figure shows the path followed by the rocket



Semi major axis = R Semi minor axis $= \frac{R}{\sqrt{2}}$

Time of flight $\propto$ Area swept

$$\frac{t}{T} = \frac{\left(\frac{\pi R}{2}\cdot\frac{R}{\sqrt{2}} + \frac{R^2}{2}\right)}{\frac{\pi R^2}{\sqrt{2}}} \qquad t = T.\frac{\left(\frac{\pi R^2}{2\sqrt{2}} + \frac{R^2}{2}\right)}{\frac{\pi R^2}{\sqrt{2}}}$$

$$\left[T = \frac{2\pi}{\sqrt{GM}}a^{3/2}.\ \text{Here } a = R \qquad T = \frac{2\pi}{\sqrt{GM}}R^{3/2}\right]$$

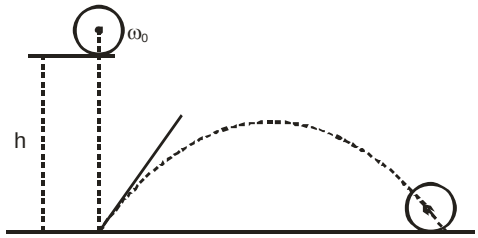
$$t = \frac{2\pi}{\sqrt{GM}}R^{3/2}\frac{\left[\frac{\pi}{2\sqrt{2}} + \frac{1}{2}\right]}{\frac{\pi}{\sqrt{2}}} \qquad t = \frac{R^{3/2}}{R\sqrt{\frac{GM}{R^2}}}\left[\pi + \sqrt{2}\right]$$

$$t = \sqrt{\frac{R}{g}}\left[\pi + \sqrt{2}\right]$$

18. **4**

Sol. v_{cm} just before striking $= \sqrt{2gh}$

v_{cm} just after striking $= e\sqrt{2gh}$



Impulse: $\int N\,dt = m(1+e)\sqrt{2gh}$

Frictional impulse in horizontal direction

$\mu\int N\,dt = \mu.m(1+e)\sqrt{2gh} \Rightarrow mv_x = \mu m(1+e)\sqrt{2gh}$

$\Rightarrow v_x = \mu(1+e)\sqrt{2gh}$

Time of flight $= \frac{2u_y}{g} = 2\times e\frac{\sqrt{2gh}}{g}$

Range $= \frac{\mu(1+e)\sqrt{2gh}\times\sqrt{2gh}.e}{g}$

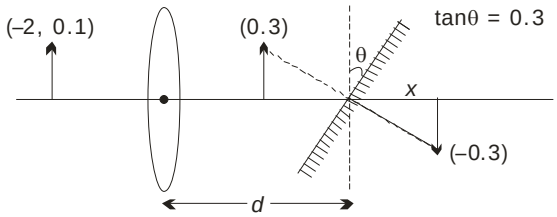
$= 2\mu.(1+e).e.2h = 4\mu(1+e)eh$

19. **5**

Sol. $\frac{1}{u} - \frac{1}{v} = \frac{1}{f} \qquad \Rightarrow \frac{1}{u} + \frac{1}{2} = \frac{10}{15}\frac{2}{3}$

$\Rightarrow \frac{1}{u} = \frac{2}{3} - \frac{1}{2} = \frac{1}{6} \Rightarrow u = 6m$

$m = \frac{u}{v} = -3$



$\tan\theta = \frac{0.3}{x} \qquad \Rightarrow 0.3 = \frac{0.3}{x} \qquad \Rightarrow x = 1$

$x = 6 - d \qquad \Rightarrow 6 - d = 1$

Ans.: $d = 5m$

20. **4**

Sol. $d\phi = \left(\frac{\mu_0 i}{2\pi x}\right)\cdot\left(\frac{b}{h}(x-a)dx\right)$

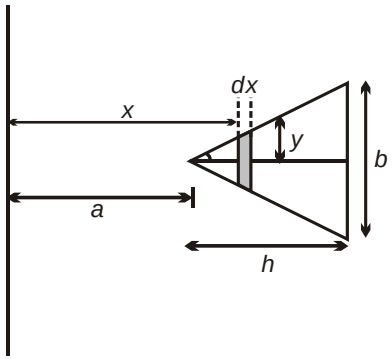
$$\int d\phi = \int_a^{a+h} \frac{\mu_{0i}}{2\pi}\left[\frac{b}{h}\left[1-\frac{a}{x}\right]\right]dx$$

$$\phi = \frac{\mu_0 i}{2\pi}\frac{b}{h}[h - a\ln x]_a^{a+h}$$

$$\phi = \frac{\mu_0 i}{2\pi}\frac{b}{h}\left[h - a\ln\left(\frac{a+n}{a}\right)\right] \qquad \phi = \frac{\mu_0 i}{2a}\frac{b}{h}[h - a\ln 2]$$

$$M = \frac{\phi}{i} \Rightarrow M = \frac{\mu_0}{2\pi}\frac{b}{h}[h - a\ln 2] = \frac{4\pi\times10^{-7}}{2\pi}\times 2\times[1-\ln 2]$$

Ans.: $= 4\times10^{-8}[1-\ln 2]$



SECTION – D

21. **00000.15 m**

Sol. $\frac{1}{2}mv^2 = \left(\frac{hc}{\lambda}-\phi\right) \Rightarrow v = \sqrt{\frac{2}{m}\left(\frac{hc}{\lambda}-\phi\right)}$

$$r = \frac{mv}{qB} \Rightarrow r = \frac{m\sqrt{\frac{2}{m}\left(\frac{hc}{\lambda}-\phi\right)}}{qB}$$

Ans.: r = 0.15 m.

22. **00126.00 °C**

Sol. Heat generated with in element = heat conduct by the element.
Let heat generated per unit volume per unit time be, x

$$x\times\pi r^2.l = -k.2\pi r l\frac{dT}{dr} \qquad \int_0^R x\,r\,dr = -\int_{T_C}^{T_S} 2k\,dT$$

$$x.\frac{R^2}{2} = -2k\left[T_S - T_C\right] \Rightarrow x\frac{R^2}{2} = 2K\left[T_C - T_S\right]$$

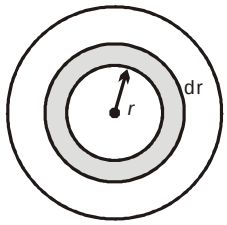
$$T_C - T_S = \frac{x}{2k}\frac{R^2}{2}$$

$$\Rightarrow T_C = T_S + \frac{x}{2k}\frac{R^2}{2}$$

$$= T_S + \left(\frac{P}{\pi R^2 l}\right)\times\frac{R^2}{2}\frac{1}{2k} = T_S + \frac{P}{4Alk}$$

$$= T_S + \frac{2\times10^3}{\pi\times4\times0.5\times15} \qquad T_C = 105 + 21$$

Ans.: $T_C = 126^0 C$



23. **00001.32 kg**

Sol. $2\cdot{}_1D^2 \longrightarrow {}_1T^3 + {}_1p^1 + \Delta E$

$\Delta m = [2\times 2.01458 - (3.01605 + 1.00728)]$amu $\quad \Delta m = 0.00583$ amu

$E = \Delta mc^2 = 8.71\times 10^{-13} J$

$8.71\times 10^{-13} J$ ____ 2 molecules

1 J ____ $\dfrac{2}{8.71\times 10^{-13}}$ molecules

$10^9 J$ ____ $\dfrac{2}{8.71\times 10^{-13}}\times 10^9$ molecules $= 2.2962\times 10^{21}$ molecules

No. of molecules is used in a day $= 2.2962\times 10^{21}$ × 86400 × 2 (efficiency 50%)

$= 396783.36\times 10^{21}$ molecules

Therefore mass used in a day $= 1.32$ kg **Ans.**

Chemistry

PART – II

SECTION – A

24. **A**

Sol. $FeSO_4 + aq. \longrightarrow FeSO_4 + 110.52$

$FeSO_4 \cdot 7H_2O + aq. \longrightarrow FeSO_4 + 7H_2O - 6.62$

$- \quad - \quad - \quad -$

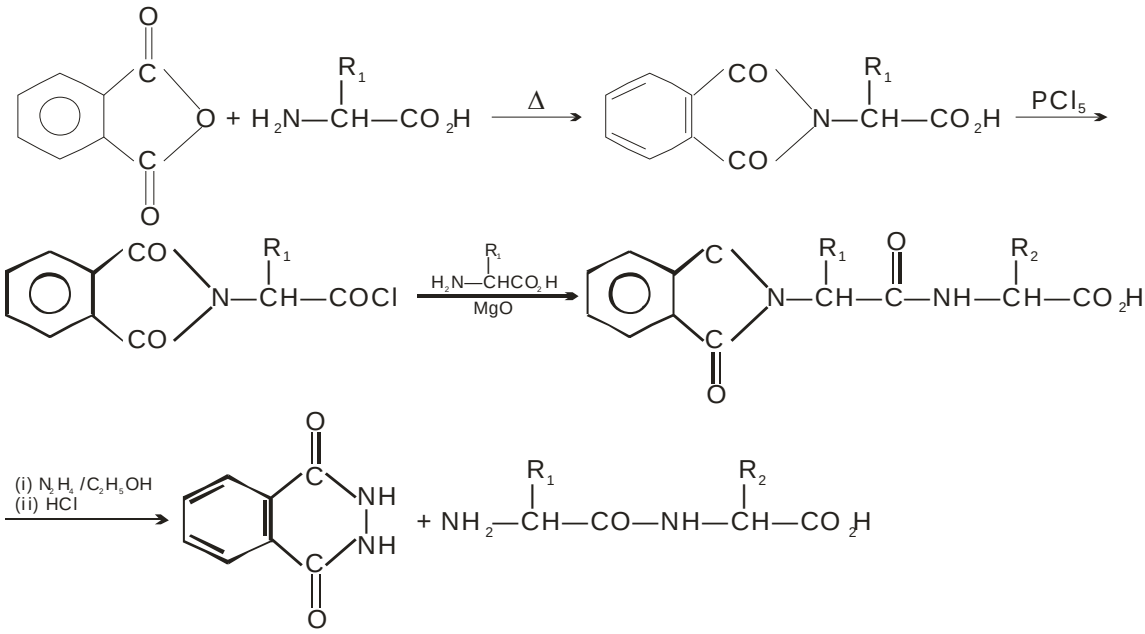
$FeSO_4 + 7H_2O \longrightarrow FeSO_4 \cdot 7H_2O + 117.14$

25. **B**

Sol. [Fact]

26. **A**

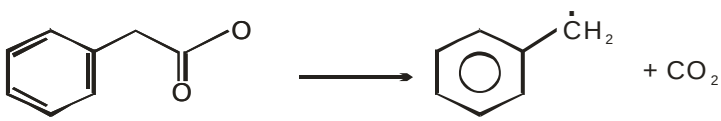
Sol.



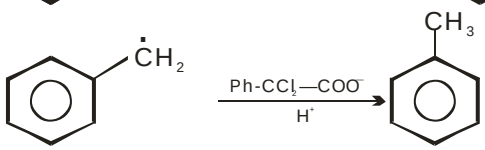
27. **BC**

Sol.

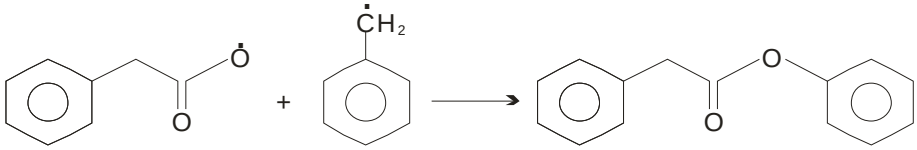
Step-1



Step-2



Step-3



28. **BC**

Sol. $E = \frac{hc}{\lambda} = 2.075\,eV$ $\quad$ K.E. $= (2.075 - 0.4) \times 1.6 \times 10^{-19}$ Joule

$\lambda = \frac{h}{\sqrt{2m}K.E.}$

29. **ACD**

Sol. Fact

30. **BD**

Sol. $AgNO_3 + NH_4Cl \longrightarrow \underset{(X)}{AgCl} + NH_4NO_3$

$AgNO_3 + NH_5 + NaOH \longrightarrow \underset{(Y)}{[Ag(NH_3)_2]^+} + NaNO_3 + H_2O$

31. **AD**

Sol. Fact O^{2-} = weak field ligand

Fe^{2+} : octa $CFSE = -(0.4 + P_0)$ $\quad$ tetra CFSE = $-\left(\frac{3}{5} + P_0\right) = -(0.6 + P_0)$

Fe^{+3} : octa/tetra = 0 CFSE

32. **D**

Sol. Fact (experimental)

33. **B**

Sol. $E_{cell} = E^0_{cell} - \frac{0.059}{2}\log\frac{[Zn^{2+}]}{[H^+]^2}$ $\quad 0.701 = 0.76 - \frac{0.059}{2}\log\frac{(0.1)}{[H^+]^2}$

$[H^+] = 0.036$ M

Equivalud of H^+ = equivalued of NaOH = $\frac{w}{E}$

$\therefore\ w_{NaOH} = 0.036 \times 40 = 1.264$ gm

SECTION – C

34. **4**

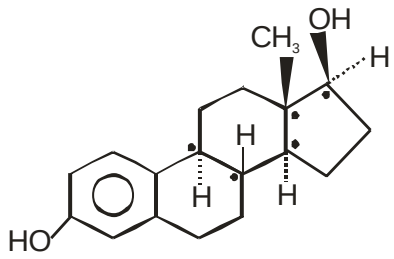
Sol. $(Na - NH_3)$, B_2, $\overset{+}{C}O$, KO_2 = para

35. **3**

Sol. $\gamma = \frac{37.2}{28.9} = 1.287 \cong 3$

36. **5**

Sol.

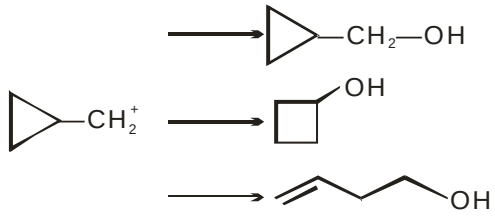


37. **2**

Sol. Fact

38. **3**

Sol.



39. **2**

Sol. Fact M_2 & M_3

40. **6**

Sol. $M_{64\%} = \frac{64 \times 1.225 \times 10}{98} = 8$ $\quad M_{20\%} = \frac{20 \times 0.98 \times 10}{98} = 2$ $\quad M_1 \longrightarrow M_2 = 6$

41. **7**

Sol. $C_xH_y \longrightarrow x = 3,\ y = 4$ Apply POAC $\quad x + y = 7$

$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$

$20 - x \quad 2(20 - x) \quad 20 - x$

$C_xH_{2y-2} + O_2 \longrightarrow nCO_2 + (n-1)H_2O$

$\left(\frac{3x-1}{2}\right) \quad x\left(\frac{3x-1}{2}\right)$

Total volume of $CO_2 \longrightarrow 20 - x + nx = 40 \quad x(n-1) = 20 \longrightarrow (i)$

Total volume of O_2 used $\longrightarrow$ 100 - 40 = 60 ml

$2(20 - x) + x\frac{(3x-1)}{2} = 60 \longrightarrow (ii)$

solve equ. (i) & (ii) find the value of *x*. $\quad x = n \quad y = 2n - 2$

42. **6**

Sol. b, d, f, h, i, j $\longrightarrow$ True

43. **5**

Sol. $K_{sp} = \left[Zn^{+2} \right] \left[S^{-2} \right]$ $S^{2-} = 10^{-15}$ $K_{H_{2s}} = \dfrac{[H^+]^2[S^{2-}]}{[H_2S]}$

$10^{-2} = \dfrac{[H^+][10^{-15}]}{10^{-4}}$ $H^+ = 10^{-5}$ $pH = 5$

SECTION – D

44. **00100.00 KJ**

Sol. $E_a = \dfrac{K_1 Ea_1 + K_2 Ea_2}{K_1 + K_2}$

45. **00000.66**

Sol. $\pi_1 = \pi_2$ $iC_1 RT = C_2 RT$ $i = 7/3$

$MgCl_2 \longrightarrow Mg^{2+} + 2Cl^-$ $i = 1 + 2\alpha$ $\alpha = 0.66$

46. **00009.87**

Sol. $I.P = \dfrac{10.79 + 8.95}{2} = 9.87$

Mathematics PART – III

SECTION – A

47. **B**

Sol. $2\cos^4 x - 2\cos^3 x + \cos^2 x + \cos x - 1 = 0$

$\cos x = t$

$2t^4 - 2t^3 + t^2 + t - 1 = 0$

$\Rightarrow \quad 2t^4 - 2t^3 + 2t^2 - t^2 + t - 1 = 0$

$\Rightarrow \quad 2t^2(t^2 - t + 1) - 1(t^2 - t + 1) = 0$

$\Rightarrow \quad (\cos^2 x - \cos x + 1)(2\cos^2 x - 1) = 0$

$\Rightarrow \quad \cos^2 x = 1/2 \quad \Rightarrow \quad \sin^2 x = 1/2$

48. **A**

Sol. $\int \frac{x^2 \sin 2x - \sin 2x(x \tan x - 1)}{(x^2+1)^2}\, dx$

$= \int \frac{(x^2+1)\sin 2x - 2x\sin^2 x}{(x^2+1)^2}\, dx$

$= \int \frac{d}{dx}\left(\frac{\sin^2 x}{x^2+1}\right) dx$

$= \frac{\sin^2 x}{x^2+1}$

49. **B**

Sol. Put y = ux and solve.

50. **BC**

Sol. Hyperbola if a = 1, parabola if $a = 2(1+\sqrt{2})$

51. **CD**

Sol. $a_n = a_{n-1} + 2a_{n-2}; n > 2$, $a_n = 2^{n-1}$

52. **AD**

53. **ACD**

Sol. $Z\bar{Z} + 2\sqrt{2}\,|Z - 1| - 4 = 0 \quad Z\bar{Z} - \sqrt{2}\bar{Z}i + \sqrt{2}iZ = 0$

$\Rightarrow \quad |Z+1| + |Z-1| = 2\sqrt{2} \quad \Rightarrow \quad |Z - \sqrt{2}i| = \sqrt{2}$

$\Rightarrow \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad x^2 + (y - \sqrt{2})^2 = 2$ .....(2)

$\Rightarrow \quad a = \sqrt{2}$ & $b = 1$ .....(1)

(2) implies $x = \sqrt{2}\cos\theta, \quad y = \sqrt{2}(1 + \sin\theta)$

then maximum & minimum value of (1)

$\Rightarrow \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1$

$= \quad \dfrac{(\sqrt{2}\cos\theta)^2}{(\sqrt{2})^2} + \left(\dfrac{\sqrt{2}(1+\sin\theta)}{1}\right)^2 - 1$

$= \quad \sin^2\theta + 4\sin\theta + 2$

$= \quad (\sin\theta + 2)^2 - 2$

$m_1 = 7 \qquad m_2 = -1$

54. **ABD**

Sol. $N(-2,2)$

$\angle NOM = 45^0$

$ON = AN = 2\sqrt{2}$

$OA = 4$

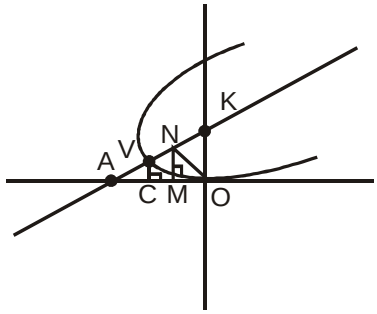
$NK = 2\sqrt{2} = 2a$ (semi L.R.)

$\Rightarrow \quad a = \sqrt{2}$

eqn. of NK : $y = x + 4$

$\therefore \quad A : (-4, 0)$

$V : (-3,1) \qquad \{\because (AV = VN)$



55. **A**

Sol. $1 + ie^{i\theta} + \dfrac{\left(ie^{i\theta}\right)^2}{2!} + ... = e^{ie^{i\theta}} = e^{-\sin\theta}(\cos(\cos\theta) +$

$i\sin(\cos\theta)$

integrate real parts.

56. **B**

Sol. Do as 55.

SECTION – C

57. **2**

Sol. $\dfrac{x}{\cos 45} = \dfrac{y}{\sin 45} = r \qquad (1)$

$y = x^2 - x + 1/x \qquad \Rightarrow \quad xy = x^3 - x^2 + 1$

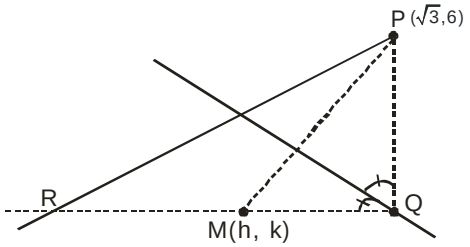
use (1) to find cubic in r.

58. **1**

Sol. Sketch the graph

59. **2**

Sol.



M(h, k) is fixed.

60. **1**

61. **8**

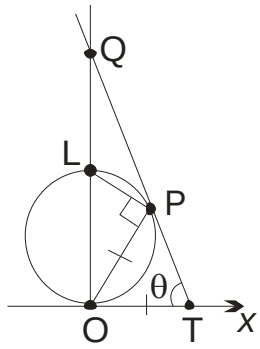
Sol. ΔOPQ

$$\frac{OQ}{\sin(\pi-\theta)}=\frac{OP}{\sin\left(\frac{\pi}{2}-\theta\right)}$$

$OQ = OP\tan\theta$ (1)

$$\frac{OP}{4}=\sin 2\theta$$

$\Rightarrow \quad OP = 4\sin 2\theta$ (2)



From (1), (2)

$OQ = 8\sin^2\theta$;

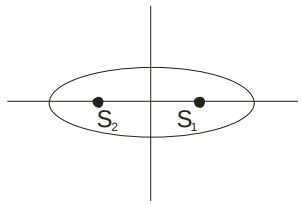
$\lim\limits_{\theta\to\pi/2} OQ = 8$

62. **9**

63. **1**

Sol. $(2x-y-5)+\lambda(x-2y-1)=0$ passes through S_1

& $(3x-y+4)+\mu(x-2y+3)=0$ passes through S_2



64. **5**

65. **0**

Sol. $F'(x)=f(x)\Rightarrow c\,f(x)\geq f'(x)$

$\Rightarrow \quad e^{cx}\frac{d}{dx}\left(e^{-cx}f(x)\right)\leq 0 \quad \Rightarrow \quad e^{-cx}f(x)$

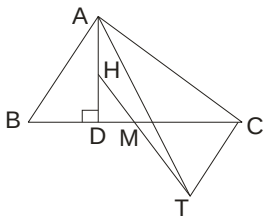
is decreasing in.

$\Rightarrow$ $f(x)$ is decreasing function but $f(0)=0$ & it is non-negative given.

$\Rightarrow$ $f(x)=0$

66. **4**

Sol.



SECTION – D

67. **00002.63**

Sol. $\lambda=\left(\frac{1}{19}-\frac{1}{20}\right)$

$\therefore \quad 1000\lambda=\frac{1000}{19\times 20}=2.631$

68. **00000.50**

Sol. $$\int \frac{(5x^4 + 3x^2)dx}{x^{10} + 2x^8 + x^6 + 2x^5 + 2x^3 + 2}$$

$$= \int \frac{5x^4 + 3x^2}{(x^5 + x^3 + 1)^2 + 1} dx$$

69. **00121.17**

Sol. $x^3 dy = (3x^2 y + x^2 + x^7)dx$

$$\Rightarrow \quad (x^3 dy - 3x^2 y\, dx) \quad = (x^2 + x^7)dx$$

$$\Rightarrow \quad \frac{x^3 dy - 3x^2 y}{(x^3)^2} \quad = (x + x^{-4})dx$$

$$\Rightarrow \quad d\left(\frac{y}{x^3}\right) \quad = (x + x^{-4})dx$$